

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/228543697>

Polyelectrolyte–macroion complexation. I. Effect of linear charge density, chain length, and macroion charge

Article in *The Journal of Chemical Physics* · August 2001

DOI: 10.1063/1.1385792

CITATIONS

141

READS

239

2 authors, including:



Marie Skepö

Lund University

97 PUBLICATIONS 2,380 CITATIONS

SEE PROFILE

Polyelectrolyte–macroion complexation. I. Effect of linear charge density, chain length, and macroion charge

Marie Jonsson^{a)} and Per Linse

Physical Chemistry 1, Center for Chemistry and Chemical Engineering, Lund University,
P.O. Box 124, S-221 00 Lund, Sweden

(Received 21 September 2000; accepted 24 May 2001)

The complexation between a linear flexible polyelectrolyte and one or several oppositely charged macroions was examined by employing a simple model system with focus on the electrostatic interactions. The composition and the structure of the complex as well as conformational data of the polyelectrolyte were obtained by using Monte Carlo simulations. These properties were investigated at different linear charge densities of the polyelectrolyte, different chain lengths of the polyelectrolyte, and different macroion charges, all at different numbers of macroions at constant volume. The binding isotherms obtained are Langmuir type, and in excess of macroions the polyelectrolyte–macroion complex displays a charge reversal. Upon complexation, the polyelectrolyte extension first reduces and thereafter increases as the number of complexed macroions increases, the minimal extension appearing for a neutral complex. Macroions prefer to complex to central polyelectrolyte segments, but for a neutral or an overcharged complex the entire polyelectrolyte chain is involved. In the complex, the macroions are able to come into direct contact with each other, despite their mutual electrostatic repulsion. The distribution of the small ions is also presented. © 2001 American Institute of Physics. [DOI: 10.1063/1.1385792]

I. INTRODUCTION

Aqueous solutions of charged macromolecules such as colloids, polyelectrolytes, and mixtures of charged colloids and polyelectrolytes have gained large scientific attention in the last several decades for both fundamental and practical reasons. Nearly all industrial and biological processes involve solutions of charged macromolecules, i.e., paint, detergents, drug delivery, and cosmetics, but also many diseases are associated with malfunctions at the colloidal level. The physiochemical properties in those systems are dominated by electrostatic forces, which are still not fully understood and need to be further characterized.

Aqueous solutions containing polyelectrolytes and self-associating charged surfactants forming micelles constitute an important class of such systems that have been studied experimentally for over 50 years. Several review articles have appeared over the years^{1–4} and the more recent work has also been summarized.⁵ These and later investigations^{6–8} have involved many different techniques resulting in a fairly well understood phase behavior, and an understanding of the organization on the molecular level is now emerging. In the dilute regime, the prevalent view is that the so-called “bead-and-necklace” model well describes the complex formed between charged surfactants and water solute polymers, in particular opposite charged polyelectrolytes. Hence, at sufficiently low surfactant concentration at which no micelles exist, the interaction between the polymer and a surfactant molecule resembles that between the polymer and a simple ion. At higher surfactant concentrations, at which surfactant micelles appear at the present conditions, a complex

is formed between polymers and micelles with the polymer wrapping the micelle but not penetrating it. The micelles resemble closely those existing in the corresponding polymer-free system. For more hydrophobic linear polymers and water-soluble polymers with hydrophobic side chains, the structure of the complex is, however, different. A biological example is the association of DNA with oppositely charged octamers of histone proteins to form a nucleosome bead. Nucleosome beads are connected by DNA linkers in a “beads-and-necklace” structure.

The polymer–surfactant interaction has also been subjected to a considerable theoretical interest. The recognition of strong hydrophobic interactions among the surfactants, leading to the surfactant-centered view rather than to the polymer-centered one, started to affect the theoretical treatment ~15 years ago. Initially, the complexation was analyzed employing thermodynamic approaches,^{9–11} whereas later simulation techniques,^{12–15} and lattice mean-field theory¹⁶ have been used. Many of these advances have recently been summarized.¹⁷ In the previous model simulations,^{12–15} it was shown that in the polyelectrolyte–macroion complex the polyelectrolyte might be wrapped around the macroion while the tails are stretching away. In particular, the influence of the flexibility of the polyelectrolyte,¹² the linear charge density of the polyelectrolyte,¹³ and the surface charge density of the macroion¹⁴ on the structure of the complex and on the thermodynamics of the complexation process were investigated. Subsequently, the complexation between one macroion and several oppositely charged polyelectrolytes was considered.¹⁵

The adsorption of polyelectrolytes onto charged curved

^{a)}Electronic mail: marie.jonsson@fkem1.lu.se

surfaces in general has also been examined by theoretical means. By using a variational approach and a ground state approximation, density profiles of adsorbed polymer segments were determined as a function of characteristic parameters describing the system,¹⁸ whereas other contributions originating from different assumptions were focused on the possibilities to overcharge a spherical macroion by an oppositely charged polyelectrolyte.^{19–21} Recently, more detailed molecular models have been employed to investigate the effect of added salt^{22,23} and of the bare polyelectrolyte stiffness²² and the case of complexation between one polyelectrolyte and several macroions has also been theoretically addressed very recently.^{24,25} In recent simulation studies, the influence of the radius of the macroion on the polyelectrolyte adsorption²¹ and of the salt on the structure of the polyelectrolyte-macroion complex²³ were investigated.

Also, solutions of macroions with short flexible charged oligomers as counterions have been studied by simulations.²⁶ Two types of system instabilities were found, (i) one arising from the macroions being bridged by the flexible counterions and (ii) one by an effective attraction between macroions arising from spatial correlations between different oligomer charges residing on different macroions. The latter is of the same type as appearing in systems with simple counterions.^{27,28}

The aim of the present study is to extend the model simulations of the complexation between polyelectrolytes and oppositely charged macroions in aqueous solution to systems involving *more than one* macroion. We thus deal with a system comprised of one single linear polyelectrolyte and one or several oppositely charged macroions with their counterions in a finite volume. The composition and the structure at different linear charge densities of the polyelectrolyte, different chain lengths of the polyelectrolyte, and different macroion charges have been investigated at different numbers of macroions present. Most noticeable is that in excess of macroions the polyelectrolyte becomes consistently overcharged and that the macroions might come into contact with each other despite their strong mutual electrostatic repulsion. The focus of our study is closely related to the theoretical ones by Schiessel *et al.*²⁴ and by Nguyen and Schklovskii.²⁵

The outline of this paper is as follows: Section II describes the model and some simulation aspects. Our results are given in Sec. III, whereas Sec. IV contains a discussion of some of our findings. The paper ends with a summary given in Sec. V.

II. METHOD

A. Model

The solutions of the polyelectrolyte and the varied amount of charged macroions are described using the primitive model. In this approach, charged hard spheres differing in charge and in size represent the ionic particles, whereas the solvent enters the model only through its relative dielectric permittivity. The model contains four different types of charged particles: (i) connected spheres representing a hydrophilic linear polyelectrolyte, (ii) large unconnected spheres representing the macroions, (iii) small unconnected

spheres representing small positive ions (cations), and (iv) small unconnected spheres representing small negative ions (anions). The polyelectrolyte is described as a freely jointed chain of the charged hard spheres (segments) connected by harmonic bonds with its intrinsic chain flexibility regulated by harmonic angular energy terms.

The total potential energy of the system can be expressed by

$$U = U_{\text{nonbond}} + U_{\text{bond}} + U_{\text{angle}}. \quad (1)$$

The nonbonded energy is pairwise additive according to

$$U_{\text{nonbond}} = \sum_{i < j} u_{ij}, \quad (2)$$

and for pair ij , where i and j denote either a polyelectrolyte segment (seg), a macroion (M), a cation, or an anion, it is given by

$$u_{ij}(r_{ij}) = \begin{cases} \infty, & r_{ij} < R_i + R_j \\ \frac{Z_i Z_j e^2}{4\pi\epsilon_0\epsilon_r} \frac{1}{r_{ij}}, & r_{ij} \geq R_i + R_j, \end{cases} \quad (3)$$

where Z_i is the charge of particle i , R_i the radius of particle i , e the elementary charge, ϵ_0 the dielectric permittivity of vacuum, ϵ_r the relative dielectric permittivity of water, and $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ the distance between the centers of particles i and j with $\mathbf{r}_i$ denoting the position of particle i . The dielectric permittivity is constant throughout the system, and hence surface polarization is neglected.

The remaining two terms in Eq. (1) are the bond and angular energy potentials, which apply only to the polyelectrolyte. The harmonic bond energy is given by

$$U_{\text{bond}} = \sum_{i=1}^{N_{\text{seg}}-1} \frac{k_{\text{bond}}}{2} (r_{i,i+1} - r_0)^2, \quad (4)$$

where $r_{i,i+1}$ denotes the distance between two connected segments with the equilibrium separation $r_0 = 5 \text{ \AA}$ and the force constant $k_{\text{bond}} = 0.4 \text{ N/m}$, and where N_{seg} denotes the number of segments of the polyelectrolyte. The angular energy is represented by

$$U_{\text{angle}} = \sum_{i=2}^{N_{\text{seg}}-1} \frac{k_{\text{ang}}}{2} (\alpha_i - \alpha_0)^2, \quad (5)$$

where α_i is the angle formed by the vectors $\mathbf{r}_{i+1} - \mathbf{r}_i$ and $\mathbf{r}_{i-1} - \mathbf{r}_i$ made by three consecutive segments with the equilibrium angle $\alpha_0 = 180^\circ$ and the force constant k_{ang} . In this study we have focused on intrinsically flexible chains only, hence $k_{\text{ang}} = 0$, whereas in subsequent contributions $k_{\text{ang}} > 0$ will be considered.²⁹ In addition to the angular potential, the electrostatic interaction among the segments also contributes to the rigidity of the polyelectrolyte, and this electrostatic contribution is of course affected by other ionic particles present.

We have considered four different systems with different combinations of polyelectrolyte and macroion properties. In system I, the polyelectrolyte has $N_{\text{seg}} = 40$ segments each with one elementary charge, $Z_{\text{seg}} = 1$, making the polyelectrolyte charge $Z_p = N_{\text{seg}} Z_{\text{seg}} = 40$. The macroion has a radius of $R_M = 15 \text{ \AA}$ and possesses ten negative elementary charges.

TABLE I. General data of the model.

Box length	$L = 257.92 \text{ \AA}$
Macroion radius	$R_M = 15 \text{ \AA}$
Segment radius	$R_{\text{seg}} = 2 \text{ \AA}$
Small ion radius	$R_{\text{ion}} = 2 \text{ \AA}$
Macroion charge	$Z_M = -10, -20$
Segment charge	$Z_{\text{seg}} = 0.5, 1$
Small ion charge	$Z_{\text{ion}} = \pm 1$
No. of segments	$N_{\text{seg}} = 20, 40$
No. of macroions	$N_M = 0-16$
No. of cations	$N_{\text{cation}} = 0-320$
No. of anions	$N_{\text{anion}} = 20, 40$
Temperature	$T = 298 \text{ K}$
Relative dielectric permittivity	$\epsilon_r = 78.4$

In system II, the magnitude of the segment charge is reduced by a factor of 2, viz. $Z_{\text{seg}} = 0.5$, whereas in system III, the number of segments is reduced by a factor of 2, viz. $N_{\text{seg}} = 20$. Thus, the polyelectrolyte charge $Z_p = 20$ is the same in systems II and III. In system IV, the macroion charge is increased by a factor 2, viz. $Z_M = -20$. This gives a polyelectrolyte-charge to macroion-charge ratio $|Z_p/Z_M| = 4$ for systems I and 2 for systems II–IV. The counterions of both the polyelectrolyte and the macroions are monovalent and are always present, and no additional simple salt is added. The volume of the systems is kept constant throughout, and the corresponding box length $L = 258 \text{ \AA}$ is selected such that it is longer than the extended polyelectrolyte. Simulations were performed at $T = 298 \text{ K}$ and $\epsilon_r = 78.4$, and up to $N_M = 16$ macroions with their counterions were added corresponding to a macroion number density $\rho_M = 9.3 \times 10^{-7} \text{ \AA}^{-3}$ or to a macroion volume fraction $\phi_M = 0.013$. General data of the model are compiled in Table I, whereas the specifications of the four systems are given in Table II.

B. Simulation details

The equilibrium properties of the model systems were obtained by employing canonical Monte Carlo (MC) simulations according to the Metropolis algorithm.³⁰ The particles were enclosed in a cubic box and periodical boundary conditions were applied. The examination of the configurational space was accelerated by employing three different types of displacements of the polyelectrolyte: (i) translational displacement of single charged particle, (ii) pivot rotation of a part of the polyelectrolyte chain and (iii) translation of the entire chain. The probability of different trial moves was weighted to enable single-particle moves ~ 20 times more often than the other two. Initially, the particles were placed randomly in the simulation box, and after equilibrium the length of a production run was in general 1×10^6 MC passes

TABLE II. Specification of the simulated systems.

System	N_{seg}	Z_{seg}	Z_M	N_M	N_{cation}	N_{anion}
I	40	1	-10	0, 1, 2, 4, 8, 16	$10N_M$	40
II	40	0.5	-10	0, 1, 2, 4, 8, 16	$10N_M$	20
III	20	1	-10	0, 1, 2, 4, 8, 16	$10N_M$	20
IV	40	1	-20	0, 1, 2, 4, 8, 16	$20N_M$	40

(attempted moves per particle). During the production run, in the most concentrated macroion system the rms displacement of the polyelectrolyte was $225 \text{ \AA}$ and the rms displacement of the macroions $1275 \text{ \AA}$ as compared to the box length $L = 258 \text{ \AA}$.

The long-ranged Coulombic interactions were handled by using the Ewald summation technique with conduction boundary conditions.^{30,31} The compensating charge distribution surrounding charged particles was supposed to be Gaussian with a width $1/(\sqrt{2}\alpha)$, where α is a parameter at our disposal. The energy evaluations in the real and reciprocal spaces are truncated using the cutoff parameters R_{cut} and k_{cut} , respectively. By imposing the same truncation error in the two spaces, we have, to the leading order, the relation $\alpha R_{\text{cut}} = k_{\text{cut}}/(2\alpha) \equiv s$, which also defines s .^{31–33} The same analysis shows that the accuracy of a single potential energy evaluation ϵ depends on s according to $\epsilon \sim \exp(-s^2)/s^2$. Hence, for given α and s , the cutoff distance in the real space becomes

$$R_{\text{cut}} = \frac{s}{\alpha}, \quad (6)$$

and the cutoff distance in the reciprocal space

$$k_{\text{cut}} = 2s\alpha. \quad (7)$$

When R_{cut} becomes or is selected smaller than half of the box length L (here $R_{\text{cut}} \approx 125 \text{ \AA}$), the summation in the real space is restricted to the particles in the primary box only. On the contrary, $k_{\text{cut}} > 2/L$, and the summation in the reciprocal space extends over replicated boxes. In practice, a value of s is specified controlling the accuracy of the Ewald summation and α is adjusted to minimize the computational load.

In the present work we have used $s = 3.0$ throughout. By comparing the potential energies of typical equilibrium configurations using these values of R_{cut} and k_{cut} obtained from Eqs. (6) and (7), respectively, with essentially exact potential energy we obtained errors 4×10^{-6} ($N_M = 0$) and 3×10^{-5} ($N_M = 4$). These errors should be compared with the relative precision of the average potential energies 6×10^{-3} and 7×10^{-3} , respectively. Hence, the errors arising from the truncation of the Ewald summation is negligible as compared to the statistical uncertainties.

The reported uncertainty of simulated quantities is one standard deviation of the mean. It is estimated from the deviation among the means of the subdivisions of the total number of MC passes according to

$$\sigma^2(\langle x \rangle) = \frac{1}{n_s(n_s - 1)} \sum_{s=1}^{n_s} (\langle x \rangle_s - \langle x \rangle)^2, \quad (8)$$

where $\langle x \rangle_s$ is the average of quantity x from one subdivision, $\langle x \rangle$ the average of x from the total simulation, and $n_s = 10$ the number of subdivisions.

The simulations were using the integrated Monte Carlo/molecular dynamics/Brownian dynamics simulation package MOLSIM.³⁴

C. Definition of complex and charge ratios

The notion of a complex formed between the polyelectrolyte and a macroion is central. We have employed a geometrical definition, and we consider them as complexed if the distance between at least one polyelectrolyte segment and the macroion does not exceed 5 Å from their contact separation according to

$$|\mathbf{r}_{\text{seg}} - \mathbf{r}_M| \leq R_c \equiv R_{\text{seg}} + R_M + 5 \text{ Å}. \quad (9)$$

Hence, the polyelectrolyte, and even a single polyelectrolyte segment, can be involved in a complexation with more than one macroion. Since we have only one polyelectrolyte in the system, there will be at most one complex. The total number of complexed macroions in the system is denoted by N_M^c .

The concept of “stoichiometric charge ratio” (β) and the “complex charge ratio” (β^c) will be used frequently. The former is defined as the ratio between the total number of macroion charges and the total number of polyelectrolyte charges in the system, i.e.,

$$\beta \equiv |N_M Z_M / N_{\text{seg}} Z_{\text{seg}}|, \quad (10)$$

whereas the latter as the ratio of the charge of all complexed macroions and the total number of polyelectrolyte charges, i.e.,

$$\beta^c \equiv |N_M^c Z_M / N_{\text{seg}} Z_{\text{seg}}|. \quad (11)$$

Thus, the case $\beta < 1$ denotes more polyelectrolyte charges than macroion charges in the system, $\beta = 1$ an equal amount of them, and $\beta > 1$ an excess of macroion charges. Moreover, $\beta^c < 1$ implies an excess of polyelectrolyte charges in the complex (also referred to as an undercharged complex), $\beta^c = 1$ an equal amount of charges in the complex (neutral complex), and $\beta^c > 1$ an excess of macroion charges in the complex (overcharged complex).

III. RESULTS

We will now present data on the polyelectrolyte-macroion complexation for the different systems at different numbers of macroions and their counterions present in the system at fixed volume. Moreover, by comparing results of the different systems, the effect of (i) the linear charge density of the polyelectrolyte, (ii) the chain length of the polyelectrolyte, and (iii) the macroion charge on the complexation can be examined.

A. Complex composition

It is anticipated that the composition and the structure of the polyelectrolyte-macroion complex depend on the number of macroions available. Figure 1 shows typical configurations of system I for all the different numbers of macroions considered. In the absence of macroions, the polyelectrolyte displays an extended conformation, and $\sim$ half of the counterions are in the nearest vicinity of it [Fig. 1(a)]. When one macroion is present, a complex is formed where the polyelectrolyte loosely wraps around the macroion and where one or two polyelectrolyte tails are stretching away from the macroion [Fig. 1(b)]. Most of the segments appear within 10 Å from the macroion surface. With two or four macroions

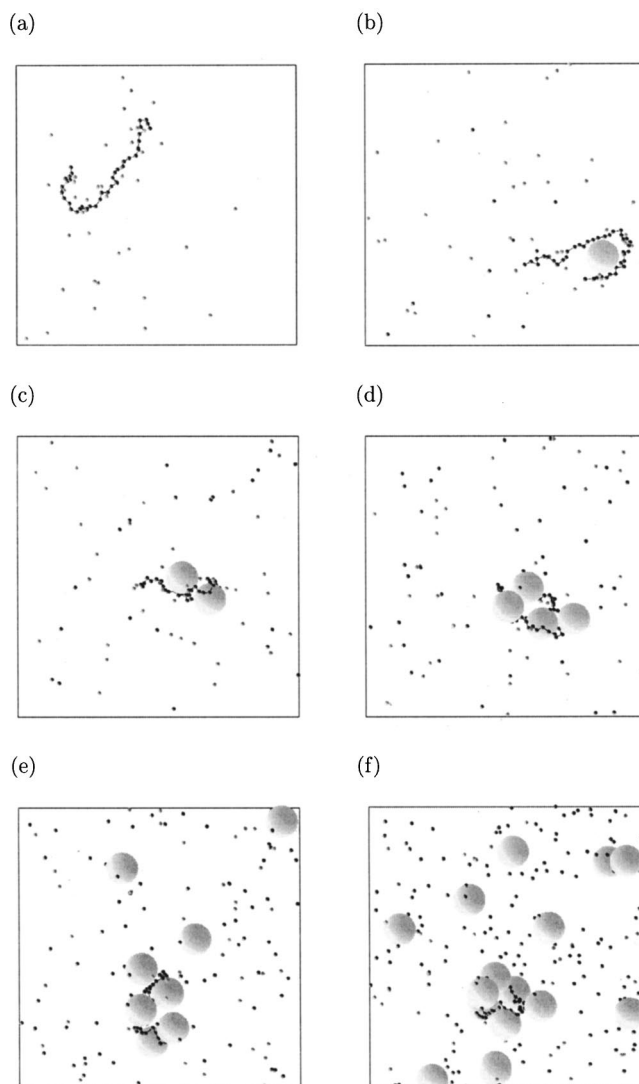


FIG. 1. Typical configurations of system I. The simulated system contains one polyelectrolyte with 40 segments (connected closed dots) and (a) $N_M = 0$, (b) $N_M = 1$, (c) $N_M = 2$, (d) $N_M = 4$, (e) $N_M = 8$, and (f) $N_M = 16$ macroions (large shaded spheres), and small positive and negative ions (unconnected dots). The box length is 258 Å, and particles are drawn to scale. Periodic boundary conditions are applied in all three directions to avoid surface artifacts.

available, all macroions are still complexed and in molecular contact with the polyelectrolyte [Figs. 1(c) and 1(d), respectively]. At even larger amount of macroions, there exist uncomplexed macroions clearly separated from the complex [Figs. 1(e) and 1(f)]. With 8 macroions present, 5 of them are complexed, and with 16, 7 of them are complexed to the polyelectrolyte. Since $N_M^c > |Z_p / Z_M| = 4$, a charge reversal is obtained, and we thus have an overcharged complex. Moreover, it appears that the polyelectrolyte becomes contracted and its tails shorter as the number of macroions in the complex increases.

These configurations also verify that the exact value of R_c for classifying a macroion as complexed with the polyelectrolyte is not important, in the present study. A macroion is either complexed with a firm contact with the polyelectrolyte or clearly separated from it. Since macroions in a complex are close to each other, strong and favorable electro-

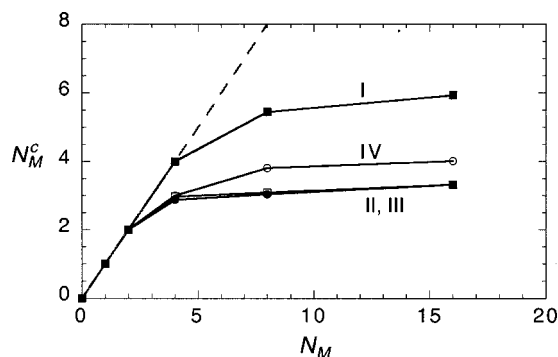


FIG. 2. Number of complexed macroions (N_M^c) vs the total number of macroions in the system (N_M) for system I (closed squares), system II (closed circles), system III (open squares), and system IV (open circles). The estimated uncertainties are smaller than the size of the symbols. The dashed line corresponds to a quantitative complexation.

static polyelectrolyte–macroion interactions are hence necessary to sustain the repulsion between them.

A representation of the number of complexed macroions as a function of the total number of macroions is referred to as a binding isotherm. Figure 2 shows the binding isotherms for the different systems. Generally, all isotherms display a Langmuir-type behavior with a strong and quantitative complexation at low N_M , a subsequent leveling off, and a plateau signaling a saturation at the higher N_M . In system I, the complexation is quantitative up to $N_M=4$ and in systems II–IV up to $N_M=2$. The plateau levels are also different among the systems.

It is reasonable to assume that the electrostatic interaction is the primary factor controlling the complex composition. The corresponding binding isotherms represented as complex charge ratio as a function of the stoichiometric charge ratio are shown in Fig. 3. In this representation, the isotherms become more alike, although some differences remain. In all systems, the complexation is quantitative up to $\beta=1$ at which the complex becomes neutral. At larger β , the isotherms start to level off and reaches a plateau value of $\beta^c \approx 1.5$ – 2 . Hence, in excess of macroions, the complex is always overcharged by 50%–100% of the polyelectrolyte charge. With these reduced parameters, systems I–III display

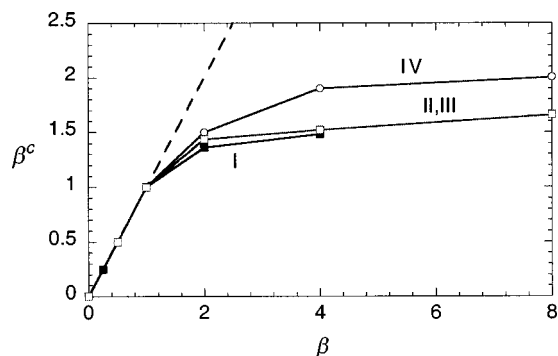


FIG. 3. Complex charge ratio ($\beta^c \equiv |N_M^c Z_M / N_{seg} Z_{seg}|$) vs the stoichiometric charge ratio ($\beta \equiv |N_M Z_M / N_{seg} Z_{seg}|$) for system I (closed squares), system II (closed circles), system III (open squares), and system IV (open circles). The estimated uncertainties are smaller than the size of the symbols. The dashed line corresponds to a quantitative complexation.

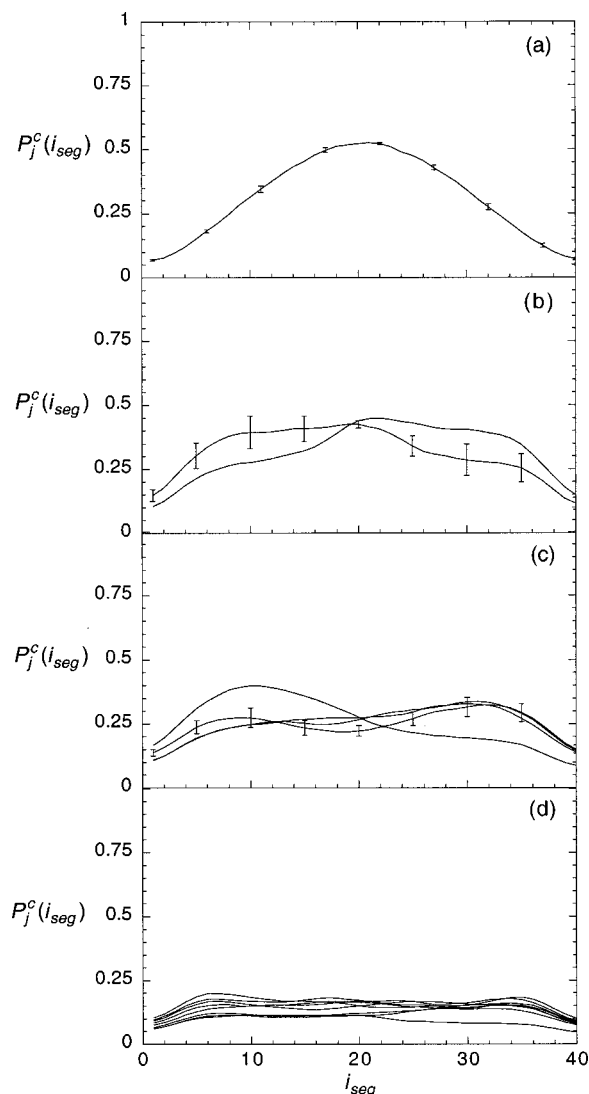


FIG. 4. Complexation probability [$P_j^c(i_{seg})$] vs segment rank (i_{seg}) for system I with (a) $N_M=1$, (b) $N_M=2$, (c) $N_M=4$, and (d) $N_M=8$ macroions. In (b)–(d), each curve denotes the complexation probability with each of the macroions present in the system. Selected uncertainty bars are given.

similar binding isotherms, whereas in system IV the overcharging capacity is larger, probably being a reflection of the stronger polyelectrolyte–macroion attraction.

B. Probability of complexed macroion

We will now examine in more detail how the polyelectrolyte is complexed to the macroions and, in particular, to which degree its different segments are involved. From the simulations the probability that segment i_{seg} is complexed to macroion j , $P_j^c(i_{seg})$, has been evaluated. Again we have used Eq. (9) as the criterion of complexation. The lower limit $P_j^c(i_{seg})=0$ implies that segment i_{seg} is never complexed to macroion j , whereas $P_j^c(i_{seg})=1$ means that the segment is always complexed to macroion j .

Figure 4 displays the complexation probability $P_j^c(i_{seg})$ for all macroions present as a function of i_{seg} for system I at $N_M=1, 2, 4$, and 8 . With one macroion present, the complexation probability is completely symmetric with respect to a reflection at $i_{seg}=(N_{seg}+1)/2$ [Fig. 4(a)]. This demonstrates

that the mobility in the configurational space is sufficient for an adequate sampling of the complexation between the macroion and the different segments of the polyelectrolyte despite the strong electrostatic macroion-polyelectrolyte interaction (≈ -40 kT in total). With two macroions present, the two complexation probabilities are within the statistical uncertainties equal [Fig. 4(b)]. Hence, during the simulation the two macroions are able to exchange position along the chain (loosely referred to as an exchange of complexation sites). With four macroions present, of which all are complexed (see Fig. 2), all complexation probabilities are nearly identical [Fig. 4(c)], and again there is a sufficient mobility in the system to be ergodic in practice. Finally, with 8 macroions, of which on average 5.5 are complexed simultaneously, nearly all $P_j^c(i_{\text{seg}})$'s are equal implying that all 8 macroions are on average complexed to a similar degree [Fig. 4(d)]. Even with 16 macroions, the same complete exchange among *all* macroions appears (data not shown). Thus, there is not only a sufficient possibility for complexed macroions to exchange complexation sites; there is also a sufficient possibility for complexed and *noncomplexed* macroions to exchange environments.

The same appearance of complete site-site as well as complexed-noncomplexed exchange also occurred in systems II and III. In system IV, however, where the macroion charge is twice as large as in system I, each individual $P_j^c(i_{\text{seg}})$ remains nearly symmetric, but for the cases with uncomplexed macroions a full exchange between complexed and uncomplexed ones was not achieved. Nevertheless, 6 of 8 and 8 of 16 macroions, of which on average 4 were complexed, were involved in the exchange. From this analysis we argue that the MC simulation protocol used is adequate for obtaining complete equilibrium results with respect to the polyelectrolyte-macroion complexation.

We will now consider the average number of macroions complexed with segment i_{seg} , $n_M^c(i_{\text{seg}})$, which is obtained by summing $P_j^c(i_{\text{seg}})$ over all macroions according to

$$n_M^c(i_{\text{seg}}) = \sum_{j=1}^{N_M} P_j^c(i_{\text{seg}}). \quad (12)$$

The complexation functions $n_M^c(i_{\text{seg}})$ for $N_M = 1, 2, 4$, and 8 macroions in each system are displayed in Fig. 5. Each curve is labeled according to the number of macroions complexed N_M^c and in parentheses the number of macroions in the system N_M . The data for $N_M = 16$ coincide with those for $N_M = 8$ and are hence not given. First, we notice that all $n_M^c(i_{\text{seg}})$ functions are symmetric with respect to a reflection at $i_{\text{seg}} = (N_{\text{seg}} + 1)/2$, as the individual $P_j^c(i_{\text{seg}})$ probabilities are, and that there is a considerable variation of the complexation functions among the different systems. In the following, we will examine these in more detail. To facilitate this examination, typical configurations are given in Fig. 6.

Consider first the complexation function for system I given in Fig. 5(a). With one macroion complexed, the complexation function $n_M^c(i_{\text{seg}})$ displays a maximum of 0.5 at $i_{\text{seg}} = (N_{\text{seg}} + 1)/2$ and minimums of 0.07 at $i_{\text{seg}} = 1$ and N_{seg} . Hence, the macroion complexes preferentially to the central part of the polyelectrolyte with two dangling tails. In the case

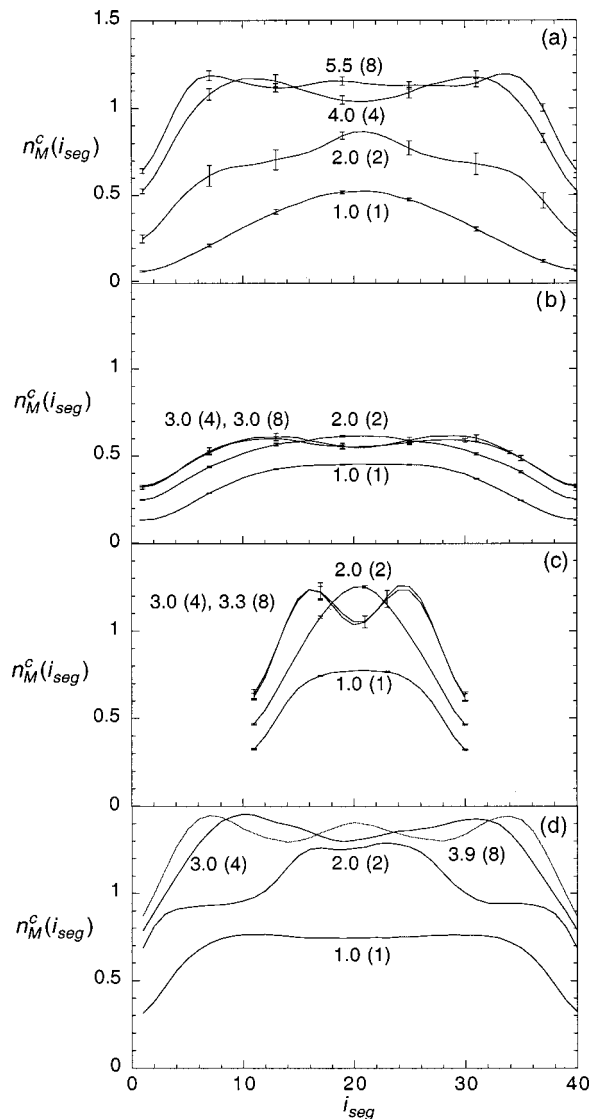


FIG. 5. Complexation function $[n_M^c(i_{\text{seg}})]$ denoting the total number of macroions complexed vs segment rank (i_{seg}) for (a) system I, (b) system II, (c) system III, and (d) system IV at indicated number of complexed macroions and in parentheses the number of macroions in the system. In (c) with $N_{\text{seg}} = 20$, the functions are centered. Selected uncertainty bars are given.

of two macroions complexed, the value of the central part of $n_M^c(i_{\text{seg}})$ is ~ 0.7 and this part extends quite far to the ends. Hence, the binding region (the part of the polyelectrolyte with high probability of complexing macroions) is larger and the dangling tails are shorter. This suggests that the macroions are mainly sequentially bounded to the polyelectrolyte as shown in Fig. 6(a). Moreover, a peculiar maximum appears at $i_{\text{seg}} = (N_{\text{seg}} + 1)/2$ indicating an enhanced probability of the ~ 5 central segments to complex macroions. Since $n_M^c < 1$, these segments are not likely to complex both macroions simultaneously, but probably they are spatially positioned between the macroions as also illustrated in Fig. 6(a). With four macroion complexed, $n_M^c(i_{\text{seg}})$ displays an even larger binding region and these segments are on average complexing one macroion. Although n_M^c still decreases as i_{seg} approaches 1 or N_{seg} , $n_M^c(1) = n_M^c(N_{\text{seg}}) \approx 0.5$ implies that there is a low probability of having significant tails as

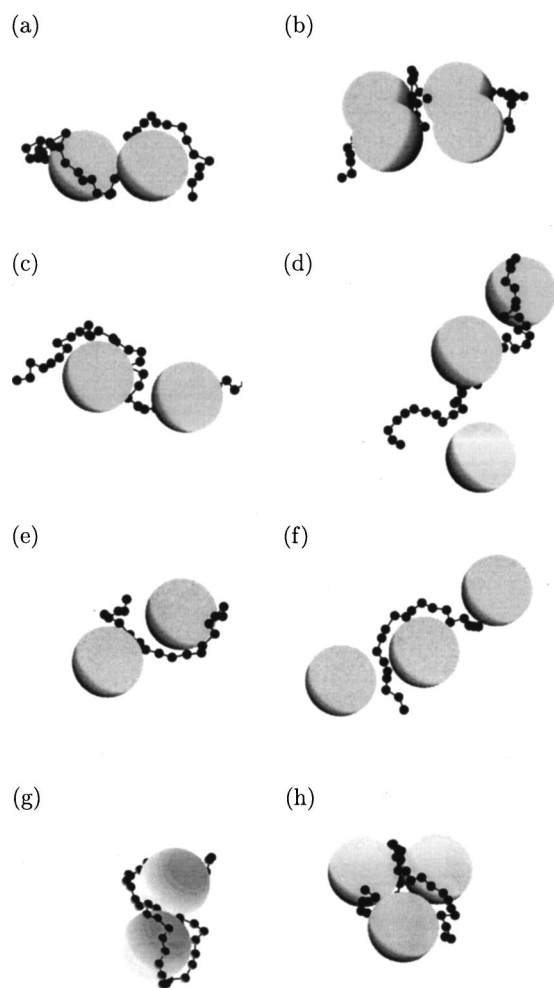


FIG. 6. Typical configurations of the polyelectrolyte-macroion complex for (a) system I with $N_M=2$, (b) system I with $N_M=4$, (c) system II with $N_M=2$, (d) system II with $N_M=4$, (e) system III with $N_M=2$, (f) system III with $N_M=4$, (g) system IV with $N_M=2$, and (h) system IV with $N_M=4$. Only the polyelectrolyte and the complexed macroions are displayed.

shown in Fig. 6(b). The central part of $n_M^c(i_{\text{seg}})$ again displays a nontrivial shape; now two maximums at $i_{\text{seg}}=10$ and 30, respectively, appear. Hence, the central segments do not any longer possess the largest probability to complex macroions. A tendency for the macroions to be grouped two and two, each group residing on each half of the polyelectrolyte chain, is suggested. Finally, with on the average 5.5 macroions complexed, the central binding region becomes broader and essentially constant, indicating on average a uniform distribution of the macroions along the polyelectrolyte. Again, only a few segments at both ends display a lower complexation probability.

The corresponding data for system II are shown in Fig. 5(b). We recall that in this system Z_{seg} has been reduced by a factor of 2 as compared to system I. The main features of $n_M^c(i_{\text{seg}})$ are similar, but the amplitudes are smaller than in system I, implying a weaker association between macroions and polymer segment. With one macroion, the complexation appears also here preferably to the central segments resulting into two tails. The central binding region becomes broader while the dangling tails becomes shorter, which corroborates

with the fact that twice as many segments are needed to neutralize the macroion charge, as compared to system I. With two macroions complexed to the polyelectrolyte, the complexation function resembles that with one macroion complexed. The shape of the complexation function is relatively smooth and the amplitude smaller than for system I, indicating a looser complex, as illustrated in Fig. 6(c). The complexation functions with three macroions complexed as obtained for $N_M=4$ and 8 coincide. The functions display the same shape as in system I with four complexed macroions, albeit with a smaller amplitude. One likely configuration is having one macroion complexed to the central part of the chain while the remaining two macroions are complexed next to it, one on each side as displayed in Fig. 6(d). Hence, a few segments between the central macroions and its two neighbors will display a larger probability to be complexed.

The complexation functions for system III are given in Fig. 5(c), where the functions are centered at $i_{\text{seg}}=10$. Here, the chain length is reduced by factor of 2 as compared to system I. With one macroion complexed, $n_M^c(i_{\text{seg}})$ displays a plateau involving ~ 10 segments, indicating that the complexation is preferred to the central segments. Of course, the polyelectrolyte tails become shorter than in system I. When two macroions are complexed, the complexation function displays a pronounced maximum for the central segments supporting the notion that both macroions are complexing the same central segments in a sandwich-type structure, see Fig. 6(e). In the saturation limit with ~ 3 macroions complexed, the shape of the complexation function changes again. Two pronounced maximums appears at $i_{\text{seg}}=6$ and 14, respectively. For this case, we anticipate a typical structure as illustrated in Fig. 6(f) with one macroion complexed to the central part of the chain and the two others on each side sharing some segments with the central macroion. A comparison of the complexation pattern between systems II and III displays similarities suggesting that the underlying structures are the same. However, the features are sharper for system III indicating more defined structures in this system. Thus, reducing the length and increasing the linear charge density of the polyelectrolyte chain at conserved total charge leads to a sharpening of the structure.

Finally, the complexation function for the remaining system is given in Fig. 5(d). Here, the macroion charge is twice as large as compared to system I. With one macroion complexed, the binding region becomes more extended as compared to system I, an obvious consequence of a larger number of segments needed to match the macroion charges. The complexation function with two macroions complexed again has a nontrivial shape. It resembles that for system I, but the central maximum is broadened with $n_M^c \approx 1.5$. Thus, the central ~ 10 segments are often simultaneously complexing both macroions as shown in Fig. 6(g). With three macroions complexed, we obtain maximums at $i_{\text{seg}}=10$ and 30 indicating a structure as illustrated in Fig. 6(h) being of the same type as in system III but less pronounced. When nearly four macroions are complexed, three maximums appear supporting a sequential binding with segments located between macroions having a significant binding to two macroions simultaneously.

TABLE III. Polyelectrolyte properties of the simulated systems where $\langle\alpha\rangle$ is the average angle between three consecutive segments, $\langle R_{ee}^2 \rangle^{1/2}$ the root-mean-square end-to-end distance, and $\langle R_G^2 \rangle^{1/2}$ the root-mean-square radius of gyration.^a

System	N_M	$\langle\alpha\rangle$ (deg)	$\langle R_{ee}^2 \rangle^{1/2}$ (Å)	$\langle R_G^2 \rangle^{1/2}$ (Å)	$\langle R_{ee}^2 \rangle / \langle R_G^2 \rangle$
I	0	122.8	126	43	8.6
	1	121.1	78	32	6.0
	2	121.4	76	30	6.4
	4	120.0	63	27	5.7
	8	119.4	71	29	6.2
	16	119.4	75	30	6.4
II	0	113.9	102	36	8.1
	1	110.8	57	26	5.0
	2	109.3	60	25	6.0
	4	108.9	61	25	6.3
	8	109.0	63	25	6.4
	16	109.0	65	26	6.6
III	0	122.0	66	23	8.2
	1	122.4	48	20	5.7
	2	119.3	45	19	6.0
	4	118.6	49	19	6.7
	8	118.6	49	19	6.8
	16	118.7	50	19	6.9
IV	0	122.8	126	43	8.6
	1	118.6	43	22	4.0
	2	120.3	64	25	6.7
	4	114.8	46	21	5.2
	8	115.3	53	23	5.7
	16	115.3	53	23	5.6

^aLargest estimated uncertainties are $\sigma(\langle\alpha\rangle)=0.1^\circ$, $\sigma(\langle R_{ee}^2 \rangle^{1/2})=2$ Å, and $\sigma(\langle R_G^2 \rangle^{1/2})=1$ Å.

Despite the fact that the complexation function only provides a fingerprint of the structure of the complex, its rich behavior demonstrates that the complexation pattern varies with the polyelectrolyte and macroion properties. The typical conformations for some of the cases given in Fig. 6 further illustrated these differences of the complexes formed.

C. Polyelectrolyte conformation

As shown in Fig. 1, the flexible polyelectrolyte seems to undergo a contraction when complexation occurs. The polyelectrolyte conformation has been characterized by (i) the average angle between three consecutive segments $\langle\alpha\rangle$, (ii) the rms end-to-end separation $\langle R_{ee}^2 \rangle^{1/2}$, (iii) the rms radius of gyration $\langle R_G^2 \rangle^{1/2}$, and (iv) the ratio $r \equiv \langle R_{ee}^2 \rangle / \langle R_G^2 \rangle$. Data for the different systems at different numbers of macroions are given in Table III.

The average angle between three consecutive segments gives information about the local stretching. For $Z_{\text{seg}}=1$, $\langle\alpha\rangle \approx 123^\circ$ is obtained independently of the chain length, and for $Z_{\text{seg}}=0.5$ we obtained $\langle\alpha\rangle=114^\circ$, demonstrating the weaker local stretching for the lower segment charge. (For the corresponding uncharged chain, $\langle\alpha\rangle=104^\circ$.) In all systems except system III, $\langle\alpha\rangle$ is reduced by 2° – 4° as the first macroion is added and reduces a further $\sim 3^\circ$ as the polyelectrolyte becomes overcharged. The initial decay of $\langle\alpha\rangle$ as N_M increases is slower in system III. Hence, the polyelectrolyte

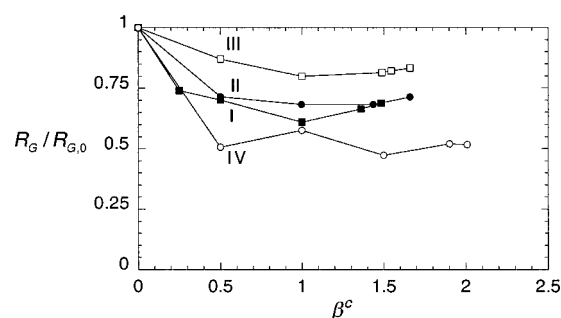


FIG. 7. Normalized radius of gyration ($\langle R_G^2 \rangle / \langle R_{G,0}^2 \rangle^{1/2}$) vs the complex charge ratio (β^c) for system I (closed squares), system II (closed circles), system III (open squares), and system IV (open circles). The quantity $\langle R_{G,0}^2 \rangle^{1/2}$ is the radius of gyration of the polyelectrolyte in the absence of macroions. The uncertainty bars are in most cases smaller than the symbols.

becomes locally more folded upon complexation. In system IV, $\langle\alpha\rangle$ displays a nonmonotonous behavior, to which we will soon return.

Both the rms end-to-end separation and the rms radius of gyration provide information on the average size of polyelectrolyte chain. The data in Table III show that both $\langle R_{ee}^2 \rangle^{1/2}$ and $\langle R_G^2 \rangle^{1/2}$ initially decrease as N_M increases, the largest drop appearing, as the first macroion is complexed.

In order to compare the polyelectrolyte contraction among the systems, Fig. 7 displays the normalized radius of gyration $\langle R_G^2 \rangle / \langle R_{G,0}^2 \rangle^{1/2}$, where $\langle R_{G,0}^2 \rangle^{1/2}$ is the rms radius of gyration of the polyelectrolyte in the absence of macroions, for the polyelectrolytes as a function of β^c . In system I, the polyelectrolyte first shrinks, reaches a minimal extension at $\beta^c=1$ (neutral complex), and then expands at $\beta^c>1$ as the number of complexed macroions increases. We anticipate that the overcharged polyelectrolyte expands primarily due to the repulsion between complexed macroions. At large β^c the original size of the noncomplexed polyelectrolyte is not yet recovered. The conformational changes in systems II and III are similar although the relative reduction of the radius of gyration is smaller (in particular for system III) as compared to system I, and the expansions in the overcharged states are weaker.

In system IV, the variation of the size of the polyelectrolyte is more irregular. After the pronounced initial reduction at $\beta^c=0.5$ ($N_M^c=1.0$), the polyelectrolyte becomes slightly more extended at $\beta^c=1$ ($N_M^c=2.0$), and then becomes contracted again at $\beta^c=2$ ($N_M^c=3.0$). A plausible explanation is that at $N_M^c=1.0$ the polyelectrolyte wraps the single macroion, at $N_M^c=2.0$ it bridges and wraps the two macroions resulting in a more extended conformation, while at $N_M^c>2$ the polyelectrolyte is again more coiled with several macroions attached [cf. Figs. 6(g) and 6(h)]. A comparison between systems III and IV, shows that a simultaneous twofold increase in the chain length and the macroion charge, conserving the polyelectrolyte-charge to macroion-charge ratio $|Z_p/Z_M|$, results in a much stronger change in the polyelectrolyte size upon complexation.

The ratio of the mean-square end-to-end distance and the mean-square radius of gyration ($r = \langle R_{ee}^2 \rangle / \langle R_G^2 \rangle$) is a quantity that characterizes the shape of the polyelectrolyte chain. In the rod-like limit $r=12$, whereas for flexible chains in

good solvent $r \approx 6.3$ and for ideal chains $r = 6$. With the exception of system IV, for the undisturbed polyelectrolyte we have $r \approx 8.5$, for neutral complexes $r \approx 6.0$, and for the overcharged ones $r \approx 6.5$ (Table III). Thus, in the absence of macroions, the polyelectrolyte chain displays an extension, although far from being a rigid rod [Fig. 1(a)]. At $\beta^c = 1.0$ it becomes similar to an ideal chain [Fig. 1(d)], and in the overcharged state it becomes more elongated again [Figs. 1(e) and 1(f)]. Thus, upon complexation the polyelectrolyte does not only change its size, i.e., contract, it also changes its shape, i.e., becomes less elongated.

D. Structure of complex macroions

The complex has further been characterized by the macroion-macroion radial distribution function (rdf) $g_{MM}(r)$, which measure the relative density of a macroion at a distance r from a given macroion. In a homogeneous solution, $g_{MM}(r)$ is conventionally defined to approach one at large distance. The macroion-macroion rdfs for the four simulated systems are given in Fig. 8. Common to all of them is a very pronounced maximum at $r \approx 30\text{--}50\text{ \AA}$ originating from the spatial correlations among the complexed macroions. As N_M is increased, the maximums become less pronounced signaling a more even distribution of the macroions in the system. When $N_M > N_M^c$, an additional broad maximum starts to appear at $r \approx 150\text{--}200\text{ \AA}$ arising from spatial correlations between complexed-uncomplexed and uncomplexed-uncomplexed macroions.

Focusing on the shorter macroion separations, we notice that complexed macroions get into molecular contact ($2R_M = 30\text{ \AA}$). With two complexed macroions, $g_{MM}(r)$ displays a maximum in the range of 32 to 37 $\text{\AA}$, and a rapid decrease at larger separation. The maximum is shifted to larger separations as more macroions enter the complex, but even when more than five macroions being complexed (system I with $N_M = 8$) the probability of finding complexed macroions separated beyond 60 $\text{\AA}$ is small. Thus, although the polyelectrolyte is linear in shape with a contour length $>200\text{ \AA}$, the complexed macroions form a compact structure as illustrated in Figs. 1(e) and 1(d). The complexed macroions are close in space, and their nearest neighbor distance is not strongly affected by N_M^c .

A comparison among the systems with two macroions complexed shows that: (i) a reduction of the segment charge leads to a larger macroion-macroion separation [cf. Figs. 8(a) and 8(b)], (ii) a reduction of the number of polyelectrolyte segments to a shortening of the macroion-macroion separation [cf. Figs. 8(a) and 8(c)], and (iii) an increase of the macroion charge leads to a *smaller* macroion-macroion separation [cf. Figs. 8(a) and 8(d)]. The width of the peaks in Fig. 8 also supports the previous notion that system II displays the loosest complex, whereas the most compact one is found in system IV.

E. Distribution of small ions

The structure of the systems has also been examined by considering rdfs between polyelectrolyte segments and the small ions which are displayed in Fig. 9 and the correspond-

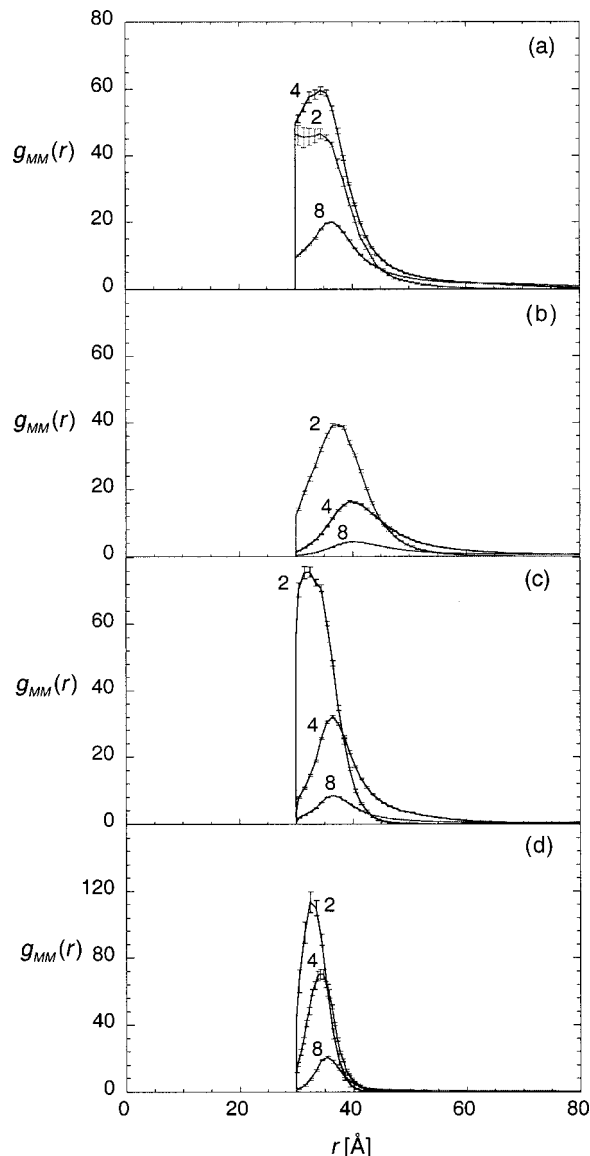


FIG. 8. Macroion-macroion radial distribution functions [$g_{MM}(r)$] at indicated number of macroions present for (a) system I, (b) system II, (c) system III, and (d) system IV.

ing ones for complexed macroions shown in Fig. 10. Results will only be presented for system I, since similar trends were obtained for the other systems.

The polyelectrolyte segment-anion (counterion to the polyelectrolyte) rdfs are shown in Fig. 9(a). In the absence of macroions, i.e., for the polyelectrolyte with its counterions only ($N_M = 0$), we have a considerable accumulation of the counterions near the polyelectrolyte beads. As the number of added macroions to the solution is increased (N_M and β^c are increasing), the accumulation of the anions near the polyelectrolyte beads is however reduced. This reduction appears to level off when the complex becomes neutral ($N_M = 4$ and $\beta^c = 1$). For the neutral complex, the integrated number of counterions within 40 $\text{\AA}$ from a polyelectrolyte segment is only 2.5. This shows that the macroions to a very large degree have replaced the small ions as counterions to the polyelectrolyte.

Figure 9(b) displays the corresponding polyelectrolyte

segment-cation (coions to the polyelectrolyte) rdf's. For undercharged complexes ($N_M < 4$ and $\beta^c < 1$), an extended region around the polyelectrolyte segments is depleted on cations. For example, at $N_M = 1$ there is less than 0.2 cations within 40 Å separation. However, for a neutral complex ($N_M = 4$ and $\beta^c = 1$), the depletion zone has decreased to ~ 40 Å and beyond this separation there is a region of considerable extension with a small excess of cations before bulk density is attained. For the overcharged complex ($N_M = 8$ and $\beta^c = 1.25$), $g_{\text{seg},\text{cation}}(r)$ display a maximum at $r \approx 35$ Å, and this spatial correlation originates from polyelectrolyte segments and cations accompanying the macroions which here carry an excess charge.

The macroion-cation (counterion to the macroion) rdf's are displayed in Fig. 10(a) and only complexed macroions are considered. For undercharged complexes ($N_M < 4$ and $\beta^c < 1$), the macroions are nearly deprived of all their small counterions. At $N_M = 1$, the integrated number of cations within 40 Å is only 0.03. For the neutral complex ($N_M = 4$ and $\beta^c = 1$), the rdf shows a fairly even radial distribution of the cations near the macroions.

Finally, Fig. 10(b) displays the macroion-anion (coion to the macroion) rdf's. For the strongly undercharged complex ($N_M = 1$), there is a strong excess of anions near the macroion, which is a consequence of the fact that the macroions do not yet fully compensate the polyelectrolyte charge. As the number of macroion increases, the accumulation of anions near the macroion decreases and for the neutral complex ($N_M = 4$ and $\beta^c = 1$), a weak excess still re-

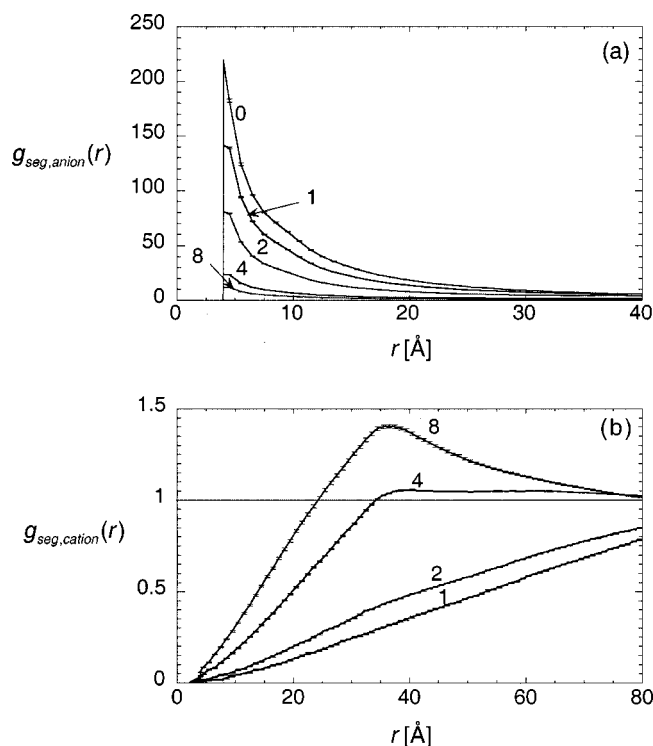


FIG. 9. (a) Polyelectrolyte segment-anion radial distribution functions [$g_{\text{seg},\text{anion}}(r)$] and (b) polyelectrolyte segment-cation radial distribution functions [$g_{\text{seg},\text{cation}}(r)$] at indicated number of macroions present for system I. The horizontal line at $g(r) = 1$ corresponds to an uncorrelated distribution.

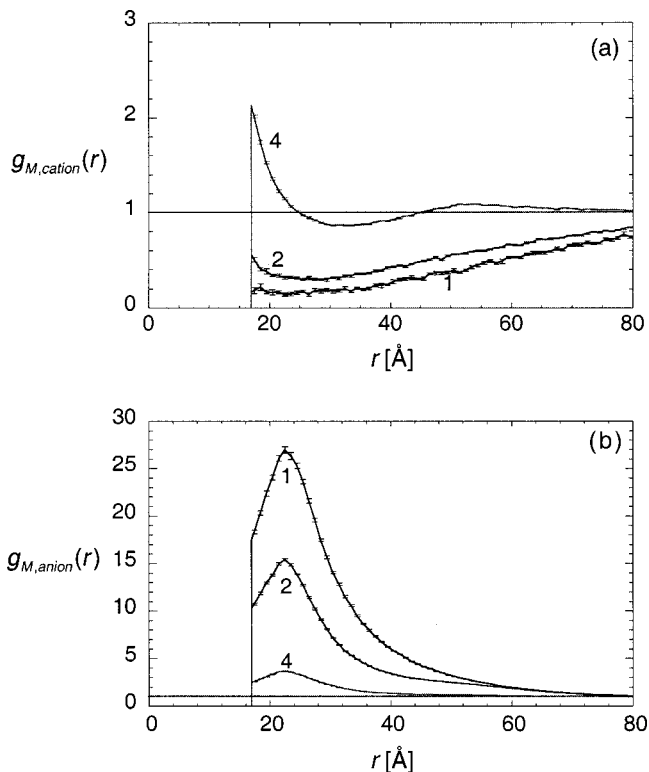


FIG. 10. (a) Macroion-cation radial distribution functions [$g_{M,\text{cation}}(r)$] and (b) macroion-anion radial distribution functions [$g_{M,\text{anion}}(r)$] at indicated number of macroions present for system I. Only complexed macroions are considered. The horizontal lines at $g(r) = 1$ correspond to uncorrelated distributions.

mains. Here, 1.3 coions are found within 40 Å from a macroion center.

Thus, these rdf's quantify the distribution of the small ions near the polyelectrolyte beads and the macroions. For undercharged complexes ($\beta^c < 1$), the rdf's confirm the accumulation of anions (counterions to the polyelectrolyte) in and near the complex, whereas for an overcharged complex ($\beta^c > 1$), there is an excess of cations (counterions to the macroions). Finally, the number of small ions in a neutral complex is small, and the average density of them in the complex and its nearest surrounding as a whole is similar to the stoichiometric one.

IV. DISCUSSION

Despite the wealth of experimental studies on the polyelectrolyte-surfactant micelle complexation, there are only a few studies focused on the structure in dilute solutions. Ilekci *et al.*⁷ have, however, shown that an addition of a sodium polyacrylate to a dilute solution of cetyltrimethylammonium bromide leads to a phase separation into two phases, where the concentrated hexagonal phase contains aligned surfactant rods and the dilute phase water and salt. By small-angle x-ray scattering, the center-to-center distance between neighboring rods was determined to be ~ 60 Å and the rod diameter ~ 35 Å. Notwithstanding the different geometry of the colloidal particles (rods in the experiments and spheres in the model), this finding is in line with our results demonstrat-

ing that oppositely charged polyelectrolyte are able to bring oppositely charged colloidal particles close to each other.

Shimabayashi *et al.*⁶ have reported on the interaction between slightly hydrophobic polymers and charged surfactants in aqueous solution. Here, the driving force for the complexation is different from that in our model. Nevertheless, from cloud point and reduced viscosity measurements Shimabayashi *et al.* inferred that the polymer coil first shrinks and then swells as the amount of surfactant bound to the polymer increases, similar to our observations. Moreover, they attribute the expansion to an increased electrostatic repulsion between the surfactant micelles attached to the polymer, an explanation that is supported here.

Previous analytic approaches have dealt with the complexation between one polyelectrolyte and one^{19–24} or several^{24,25} oppositely charged macroions. Regarding the polyelectrolyte–macroion complexation, one aim has been to assess the possibility of the macroion to have a collapsed layer of polyelectrolyte segments carrying in total a charge Z_p^c larger than that required to neutralize the macroion, i.e., $|Z_p^c| > |Z_M|$. Note, in this context with the focus on the macroion, an overcharged complex refers to $|Z_p^c| > |Z_M|$, which differs from the previous use of the phrase.

Using a variational approach, Gurovitch and Sens¹⁹ found in the dilute limit an overcharging of $|Z_p^c|/|Z_M| = 15/6$ for point macroions. The overcharging was demonstrated to reduce as the macroion radius is increased and ceases when $R_M \approx 0.5|Z_M|e^2/(4\pi\epsilon_0\epsilon_r kT)$. Park *et al.*²⁰ also demonstrated the existence of similar overcharging by entropic arguments using the Poisson–Boltzmann equation. After the initial neutralization of the macroion by a flexible polyelectrolyte and the accompanying release of counterions, a further counterion release is achieved by *additional* adsorption of polyelectrolyte charges on the (now neutral) complex. This additional adsorption causes a further counterion release as long as the (now charged) complex has a lower charge density than the polyelectrolyte. By considering the electrostatic interaction of the complexation between a flexible polyelectrolyte and a macroion at infinite dilution at 0 K, Mateescu *et al.*²¹ also concluded the existence of an overcharged state. Moreover, they also examined a transition between a fully collapsed state and states involving one or two dangling tails. Considering the case $|Z_p| > |Z_M|$, they found for small R_M a complex formed by the macroion and the collapsed polymer segments being slightly undercharged with two polyelectrolyte tails. As R_M increases, a transition to an overcharged complex appeared and eventually a collapse of the entire polyelectrolyte on the macroion occurred. In the approach by Netz and Joanny,²² the free energy of the complex is assumed to be composed of three terms: bending energy of the polyelectrolyte chain, electrostatic attraction between the macroion and the polyelectrolyte, and electrostatic intrapolyelectrolyte repulsion, the two latter contributions described employing the Debye–Hückel theory. Within the ground state approximation (neglecting fluctuations), several complexation regimes were considered and they found a region in the (κ, Z_M) space (κ^{-1} being the Debye screening length) with $|Z_p| > |Z_M|$ and the polyelectrolyte firmly wrapping the macroion.

The present molecular model is solved essentially exactly with Monte Carlo simulations. Hence, no model simplifications as preaveraging over the small ions or linearization of the Coulomb interaction are introduced and no statistical mechanical approximations are made. The concept of overcharging is an obvious property to compare. Since there is no clear delineation between complexed and non-complexed polyelectrolyte segments in our model, we define the charge of the complexed (collapsed) polyelectrolyte segments in the polyelectrolyte–macroion complex by $Z_p^c = Z_{\text{seg}} n_{\text{seg}}$, where n_{seg} denotes the number of polyelectrolyte segments within 5 Å from a complexed macroion [see Eq. (9)]. For systems I–IV, we get $Z_p^c = 12.5, 6.7, 12.2$, and 26.8 , respectively, and the charge ratios $|Z_p^c|/|Z_M| = 1.25, 0.67, 1.22$, and 1.34 , respectively. Thus, the present study supports the notion that a macroion can be undercharged as well as overcharged by a collapsed, flexible, and oppositely charged polyelectrolyte. The present study also suggests that the overcharging increase as the linear charge density of the polyelectrolyte increases and as the macroion charge increases, but is less sensitive on the polyelectrolyte length. Moreover, since complexation functions for $N_M^c = 1$ possess a single maximum at $i_{\text{seg}} = (N_{\text{seg}} + 1)/2$, and hence are *not* bimodal, there is no evidence for a dominance of single tail configurations. This is one central aspect in several of the theoretical predictions.

The theories by Schiessel *et al.*²⁴ and by Nguyen and Shklovskii²⁵ describe the complexation appearing in a solution containing one polyelectrolyte and macroions in a similar way. Both theories are based on the assumption that the complexed macroions are individually wrapped by the polyelectrolyte chain in a solenoid conformation. After formulating a free energy expression of the complexation, the number of complexed macroions and their separation are obtained by a free energy minimization. The theories differ with respect to the free energy terms included and they were constructed to be applicable for different parts of the parameter space. The theory by Schiessel *et al.*²⁴ is adapted to a semiflexible and highly charged polyelectrolyte, where the counterions are explicitly taken into account, whereas the one by Nguyen and Shklovskii²⁵ considered the case of a total flexible and weakly charged polyelectrolyte in the absence of counterions (infinite volume). Nevertheless, their predictions are similar.

Briefly, in the case of a single macroion complexing the polyelectrolyte both theories display regimes where the macroion is overcharged by the polyelectrolyte, although the mechanisms are viewed differently. For a neutral complex, they predict that the polyelectrolyte is fully wrapping the macroions, and the distance between neighboring macroions being small. Finally, for a system with excess macroions, each complexed macroion is undercharged, or alternatively viewed the polyelectrolyte is overcharged by complexed macroions.

These over/undercharging trends are in qualitative agreement with our results. For system I, with only one macroion complexed to the polyelectrolyte, 13 of the 40 polyelectrolyte segments are complexed to the macroion giving rise to a 30% overcharged macroion. On the other hand, at the saturation limit, only 70% of the macroion charges are neutral-

ized by the polyelectrolyte, hence the macroions are undercharged. There is also an agreement that the macroions are close to each other for a neutral complex, although this appears to be the prevalent case also for over- or undercharged polyelectrolyte in our study. Although a firmer comparison among the theoretical predictions and simulation results is hampered by the different assumptions, the conformity of the results discussed indicates that these properties of the systems are robust.

V. SUMMARY

On the basis of Monte Carlo simulations of a model system representing an aqueous solution containing one polyelectrolyte and several oppositely charged macroions, the composition and the structure of the complex formed have been investigated. Despite the strong electrostatic interactions, it has been demonstrated that the MC simulation technique employed enables sufficient mobility in configurational space to make the simulations ergodic in practice.

The strong attractive electrostatic interactions between the polyelectrolyte and a macroion make the polyelectrolyte-macroion complex well defined. A macroion is either complexed with a firm contact to the polyelectrolyte or clearly separated from it. The binding isotherms are Langmuir-like, and in excess of macroions the complex becomes overcharged by 50%–100% of the polyelectrolyte charges. The charge reversal appeared for all four systems and it was strongest for the larger macroion charge. After reaching the binding plateau, the structure of the complexes was not strongly affected by uncomplexed macroions for the cases studied. When complexed, the macroions are located close to each other despite their mutual electrostatic repulsion.

A free polyelectrolyte is considerably extended by the repulsive electrostatic interaction between its segments. Upon complexation, the polyelectrolyte becomes locally less stretched and its overall shape shrinks and it becomes more spherical. The smallest extension appeared for a neutral complex and it becomes expanded again in the overcharged state, although less expanded as compared to a free polyelectrolyte.

A single macroion prefers to be complexed to the central part of the polyelectrolyte. No support of an asymmetric binding leading to tails of different lengths was detected. When the number of complexed macroions is increased, they still prefer to complex to the central part of the polyelectrolyte, but by necessity segments further away are engaged and the dangling tails become shorter. In the overcharged state, essentially no tails are left.

The comparison between properties of systems I and II shows that a reduction of the linear charge density of the polyelectrolyte leads to (i) a reduced capacity to complex macroions and (ii) a looser complex as inferred from smaller number of complexed segments and longer separation between macroions in the complex (at the same number of complexed macroions). A reduction of the polyelectrolyte length at conserved linear charge density leads to (i) a reduced capacity to complex macroions, (ii) a more compact

complex as inferred from the macroion-macroion rdfs, and (iii) a smaller contraction of the polyelectrolyte. Finally, an increase of the macroion charge results in (i) a reduced capacity to complex macroions (fewer macroions) but an increased overcharging limit, (ii) a more compact complex as deduced from the macroion-macroion rdfs and the complexation function, and (iii) a larger contraction of the polyelectrolyte.

In this study, we have employed a model containing only electrostatic and hard-sphere interactions between the polyelectrolyte, the macroions, and their counterions. Although this model is able to describe the important interactions relevant for hydrophilic polymers and surfactant micelles, the model needs to be extended for describing the complexation between less soluble polyelectrolytes or neutral polymers with surfactant micelles or other colloids. Moreover, the study has been focused on rather short polyelectrolytes as compared to real ones to make a fairly detailed examination of several systems possible. Nevertheless, we anticipate that the many aspects emerging from the present study are also valid for longer polyelectrolytes.

ACKNOWLEDGMENTS

This work was supported by grants from the Foundation of Strategic Research (SSF), through National Graduate School in Scientific Computing (NGSSC), the Swedish Research Council for Engineering Science (TFR), the Swedish National Research Council (NFR), and by computing resources by the Swedish Council for Planning and Coordination of Research (FRN) and the Center for Parallel Computers (PDC), Royal Institute of Technology.

- ¹I. D. Robb, in *Anionic Surfactants—Physical Chemistry of Surfactant Action*, edited by E. Lucassen-Reynders, Surfactant Science Series Vol. 11. (Marcel Dekker, New York, 1981).
- ²E. D. Goddard, *Colloids Surface* **19**, 301 (1986).
- ³K. Hayakawa and J. C. T. Kwak, in *Cationic Surfactants*, edited by D. N. Rubingh and P. M. Holland (Marcel Dekker, New York, 1991).
- ⁴B. Lindman and K. Thalberg, in *Interactions of Surfactants with Polymers and Proteins*, edited by E. D. Goddard and K. P. Ananthapadmanabhan (CRC Press, Boca Raton, FL, 1993).
- ⁵*Polymer-Surfactant Systems*, edited by J. C. T. Kwak, in Surfactant Science Series Vol. 77 (Marcel Dekker, New York, 1998).
- ⁶S. Shimabayashi, T. Uno, Y. Oouchi, and E. Komatsu, *Prog. Colloid Polym. Sci.* **106**, 136 (1997).
- ⁷P. Ilekki, T. Martin, B. Cabane, and L. Piculell, *J. Phys. Chem. B* **103**, 9831 (1999).
- ⁸Y. Wang, K. Kimura, Q. Huang, P. Dubin, and W. Jaeger, *Macromolecules* **32**, 7128 (1999).
- ⁹R. Nagarajan, *Adv. Colloid Interface Sci.* **26**, 205 (1986).
- ¹⁰E. Ruckenstein, G. Huber, and H. Hoffman, *Langmuir* **3**, 382 (1987).
- ¹¹Y. J. Nikas and D. Blankstein, *Langmuir* **10**, 3512 (1994).
- ¹²T. Wallin and P. Linse, *Langmuir* **12**, 305 (1996).
- ¹³T. Wallin and P. Linse, *J. Phys. Chem.* **100**, 17 873 (1996).
- ¹⁴T. Wallin and P. Linse, *J. Phys. Chem. B* **101**, 5506 (1997).
- ¹⁵T. Wallin and P. Linse, *J. Chem. Phys.* **109**, 5089 (1998).
- ¹⁶T. Wallin and P. Linse, *Langmuir* **14**, 2940 (1998).
- ¹⁷P. Linse, L. Piculell, and P. Hansson, in *Polymer-Surfactant Systems*, edited by J. C. T. Kwak, Surfactant Science Series Vol. 77 (Marcel Dekker, New York, 1998).
- ¹⁸F. von Goeler and M. Muthukumar, *J. Chem. Phys.* **100**, 7796 (1994).
- ¹⁹E. Gurovitch and P. Sens, *Phys. Rev. Lett.* **82**, 339 (1999).
- ²⁰S. Y. Park, R. F. Bruinsma, and W. M. Gelbart, *Europhys. Lett.* **46**, 454 (1999).

- ²¹E. M. Mateescu, C. Jeppesen, and P. Pincus, *Europhys. Lett.* **46**, 493 (1999).
- ²²R. Netz and J.-F. Joanny, *Macromolecules* **32**, 9013 (1999).
- ²³T. T. Nguyen and B. I. Shklovskii, *Physica A* **293**, 324 (2001).
- ²⁴H. Schiessel, R. Bruinsma, and W. M. Gelbart, *J. Chem. Phys.* (submitted).
- ²⁵T. T. Nguyen and B. I. Shklovskii, *J. Chem. Phys.* **114**, 5905 (2001).
- ²⁶J. Rescic and P. Linse, *J. Phys. Chem. B* **104**, 7852 (2000).
- ²⁷P. Linse and V. Lobaskin, *Phys. Rev. Lett.* **83**, 4208 (1999).
- ²⁸P. Linse and V. Lobaskin, *J. Chem. Phys.* **112**, 3917 (2000).
- ²⁹M. Jonsson and P. Linse, *J. Chem. Phys.* (submitted).
- ³⁰M. P. Allen and D. J. Tildesley, *Computer Simulations of Liquids* (Oxford, University Press, New York, 1987).
- ³¹D. Frenkel and B. Smit, *Understanding Molecular Simulation* (Academic, New York, 1996).
- ³²J. Kolafa and J. W. Perram, *Mol. Simul.* **1992**, 351 (1992).
- ³³V. Lobaskin and P. Linse, *J. Chem. Phys.* **109**, 3530 (1998).
- ³⁴MOLSIM, Version 3.0, P. Linse, Lund University, Sweden, 1999.